# ASSOCIATED POINTS OVER AN AFFINE OPEN ARE DETECTED ON THE MODULE OF SECTIONS BY THE CORRESPONDING PRIME

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-associated-affine-open is carried in claim_key and here rather than in a per-claim tag URL.

*Editorial note.* The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-divisors/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

*Editorial note.* Statement span 67-84 and proof span 86-117 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

*Editorial note.* The proof opens with the author's sentence that the statement also follows from a cited algebra lemma, and then says that a direct argument can be given as well; both the citation and the direct argument are preserved, and the direct argument is what is counted.

*Editorial note.* 3 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

*Allowed standard prerequisites (editorial).*

- An associated prime of a module over a ring is the annihilator of some element, and localization is exact. Used at proof lines 88-94 and 97-107. Produces the element whose annihilator is the prime in the first direction, and the element of the localization whose annihilator is the localized prime in the second.
- The stalk of a quasi-coherent module on an affine open at a prime is the localization of the module of sections at that prime, cited from the schemes chapter. Cited at proof lines 92-93. Identifies the localized module with the stalk so that association at the point follows from association at the prime.
- For a finitely generated prime ideal contained in the annihilator up to localization there is an element outside the prime killing the quotient. Used at proof lines 104-108. Upgrades association of the localized element to association of an honest element of the module of sections.

---

*Date*: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- In a locally Noetherian scheme the coordinate rings of affine opens are Noetherian, cited from the properties chapter. Cited at proof lines 112-114. Makes every prime finitely generated, which is the hypothesis of the converse direction.

*Source reference.* Original lemma algebra-lemma-associated-primes-localize: sources/stacks-divisors/pinned-original-source.tex, source lines 88–88. Same-chapter reference used in the retained author text at source line(s) 88, 6303. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma properties-lemma-locally-Noetherian: sources/stacks-divisors/pinned-original-source.tex, source lines 115–115. Same-chapter reference used in the retained author text at source line(s) 115, 164, 850. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma schemes-lemma-spec-sheaves: sources/stacks-divisors/pinned-original-source.tex, source lines 95–95. Same-chapter reference used in the retained author text at source line(s) 95. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

**Lemma 0.1.** *Let $X$ be a scheme. Let $\mathcal{F}$ be a quasi-coherent sheaf on $X$. Let $\mathrm{Spec}(A) = U \subset X$ be an affine open, and set $M = \Gamma(U, \mathcal{F})$. Let $x \in U$, and let $\mathfrak{p} \subset A$ be the corresponding prime.*

(1) *If $\mathfrak{p}$ is associated to $M$, then $x$ is associated to $\mathcal{F}$.*
(2) *If $\mathfrak{p}$ is finitely generated, then the converse holds as well.*

*In particular, if $X$ is locally Noetherian, then the equivalence*

$$\mathfrak{p} \in Ass(M) \Leftrightarrow x \in Ass(\mathcal{F})$$

*holds for all pairs $(\mathfrak{p}, x)$ as above.*

*Proof.* This follows from Algebra, Lemma algebra-lemma-associated-primes-localize. But we can also argue directly as follows. Suppose $\mathfrak{p}$ is associated to $M$. Then there exists an $m \in M$ whose annihilator is $\mathfrak{p}$. Since localization is exact we see that $\mathfrak{p}A_\mathfrak{p}$ is the annihilator of $m/1 \in M_\mathfrak{p}$. Since $M_\mathfrak{p} = \mathcal{F}_x$ (Schemes, Lemma schemes-lemma-spec-sheaves) we conclude that $x$ is associated to $\mathcal{F}$.

Conversely, assume that $x$ is associated to $\mathcal{F}$, and $\mathfrak{p}$ is finitely generated. As $x$ is associated to $\mathcal{F}$ there exists an element $m' \in M_\mathfrak{p}$ whose annihilator is $\mathfrak{p}A_\mathfrak{p}$. Write $m' = m/f$ for some $f \in A$, $f \notin \mathfrak{p}$. The annihilator $I$ of $m$ is an ideal of $A$ such that $IA_\mathfrak{p} = \mathfrak{p}A_\mathfrak{p}$. Hence $I \subset \mathfrak{p}$, and $(\mathfrak{p}/I)_\mathfrak{p} = 0$. Since $\mathfrak{p}$ is finitely generated, there exists a $g \in A$, $g \notin \mathfrak{p}$ such that $g(\mathfrak{p}/I) = 0$. Hence the annihilator of $gm$ is $\mathfrak{p}$ and we win.

If $X$ is locally Noetherian, then $A$ is Noetherian (Properties, Lemma properties-lemma-locally-Noetherian) and $\mathfrak{p}$ is always finitely generated. □